

2 A programmer uses a number of Abstract Data Types (ADT) in the programs that she is developing.

(a) A stack using memory locations 400 to 409 is used in one program.

The diagram represents the current state of the stack.

A variable `TopOfStack` pointer indicates the last value added to the stack.

Stack		
Memory location	Value	
409		
408		
407		
406		
405		
404		
403	'P'	← <code>TopOfStack</code>
402	'X'	
401	'D'	
400	'B'	

(i) Complete the answer column in the following table:

Answer	
the memory location of the value that has been on the stack for the longest time	
the number of consecutive push operations that will result in the <code>TopOfStack</code> variable containing 409	

[2]

(ii) The diagram shows the current state of the stack.

Stack		
Memory location	Value	
409		
408		
407	'A'	← <code>TopOfStack</code>
406	'C'	
405	'F'	
404	'K'	
403	'B'	
402	'S'	
401	'R'	
400	'D'	

The following sequence of operations are performed:

PUSH 'T'
POP
POP
PUSH 'Z'
PUSH 'X'
POP
PUSH 'Y'

Complete the following diagram to show the state of the stack after the operations have been performed.

Stack		
Memory location	Value	
409		
408		
407		
406		
405		
404		
403		
402		
401		
400		

[3]

(b) In a different program, the programmer decides to implement a linked list using an array of records.

(i) The array is represented in the diagram.

A pointer value of `-1` indicates the end of the linked list.

Complete the pointer field to produce a linked list that is in alphabetical order.

Index	Data	Pointer
0	"Neptune"	
1	"Jupiter"	
2	"Saturn"	
3	"Earth"	
4	"Mercury"	
5	"Uranus"	

[3]

(ii) The variable `StartPointer` contains the index of the first item in the linked list.

State the value of the variable `StartPointer`

..... [1]

(c) A third program is also being created to manage a queue.

Describe **two** features of a queue.

Feature one

.....

Feature two

.....

[2]